

Performance and Robustness Benchmarking of Battery SOC Estimation Methods for Embedded Applications

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Abstract

Robustness of battery State-of-Charge (SOC) estimation cannot be inferred from nominal accuracy alone. This study compares representatives of four major SOC-estimation classes under one causal and cell-disjoint protocol: fixed-capacity Coulomb counting (DM), SOH-adaptive Coulomb counting (HDM), a second-order equivalent-circuit observer (HECM), and a recurrent data-driven estimator (DD). LFP aging trajectories span multiple load, thermal, and SOH conditions. Performance is separated into nominal accuracy, robustness when measurements are corrupted or unavailable, and recovery after an initialization mismatch. The six-cell macro MAE under nominal inputs is 0.0690 for DM, 0.0412 for HDM, 0.0337 for HECM, and 0.0258 for DD. DD also leads several robustness cases, but no representative dominates every disturbance family and the normalized robustness ordering changes with the aggregation rule. HECM restores the paired initialization mismatch most persistently, with a mean recovery time of 1.12 h and a relapse fraction of 0.17, compared with 1.29 h and 0.78 for DD. Aggregated scores are therefore retained only as illustrative decision aids rather than a universal ranking. All isolated SOC cores reproduce their software references, remain within the measured peak runtime RAM, and meet the tested one-second STM32 deadline, although implementation costs differ

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substantially. The results demonstrate why estimator selection must combine accuracy, disturbance-specific robustness, recovery, and embedded feasibility. Conclusions apply to the tested representatives and LFP data rather than their complete model families.

Keywords: Battery management system, State of charge, Robustness benchmark, Fault tolerance, Equivalent circuit model, Recurrent neural network, Embedded implementation, Microcontroller

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase
16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate

20 *Symbols*

21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Reference SOC target at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	b_I	Multiplicative current-gain error
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 **1. Introduction**

43 *1.1. Motivation and practical relevance*

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary

lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor gain and offset errors, additive noise, initialization uncertainty after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible measurement corruption.

As summarized conceptually in Figure 1, the present benchmark treats deployment-facing estimator quality as a three-dimensional problem consisting of nominal accuracy, recovery after state inconsistency, and robustness against disturbed measurements and signal-integrity faults, embedded within broader constraints on memory and execution time. The central question is how four structurally different SOC-estimation classes, represented by the implementations studied here, respond to common failure mechanisms and microcontroller constraints, and whether their nominal ranking remains informative under those conditions. This distinction is essential because one implementation can be accurate under clean measurements yet difficult to re-anchor, sensitive to a specific input fault, or too costly under its trained inference semantics.

1.2. State of the art in battery state estimation

The literature commonly groups battery-state estimators into direct-measurement or integration-based approaches, model-based approaches based on equivalent-circuit or electrochemical models, hybrid approaches that combine physical structure with learned components, and purely data-driven models based on neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically based on Coulomb counting, are attractive because they are simple, computationally inexpensive, and easy to deploy. However, they remain structurally sensitive to current-gain error, missing data, and initialization errors because

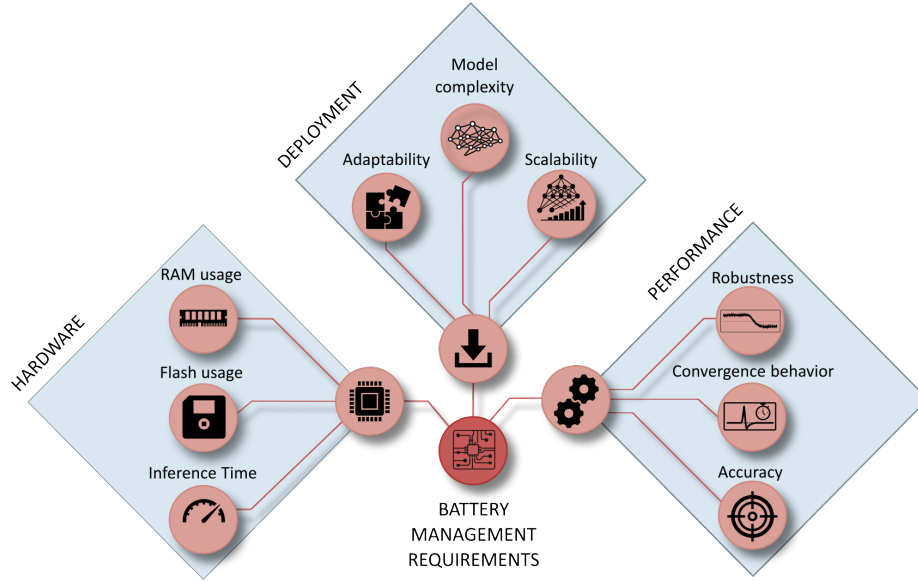


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

they lack an internal correction mechanism [7, 4]. Model-based approaches, especially ECM- and observer-based estimators, offer physical interpretability and voltage-based correction capability, but their practical behavior depends strongly on the quality of model parametrization, operating-point observability, and measurement quality [8, 2, 4].

Hybrid estimators seek to combine physical state propagation with learned adaptation of capacity, parameters, or auxiliary states, which has made joint SOC/SOH estimation an active research direction [9, 10, 11]. This combination can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented imple-

90 mentations, pruning, and quantization [12, 13, 14, 15, 3].

91 *1.3. Performance and embedded deployment gap*

92 A large share of the SOC estimation literature still emphasizes clean-data ac-
93 curacy, average error metrics, or computational efficiency, while comparatively
94 fewer studies analyze how estimators behave under realistic measurement faults
95 and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain
96 confined to a single model family, for example observer-based robustness against
97 modeling and measurement errors [4] or robust estimation concepts for specific
98 BMS architectures under noise and uncertainty [5], whereas cross-family com-
99 parisons under the same disturbance protocol and under comparable embedded
100 constraints remain limited. Benchmarking studies in the broader BMS con-
101 text underline the importance of reproducible scenario design and deployment-
102 relevant validation, but they do not by themselves close the gap between clean-
103 data estimator comparison and measurement-level performance benchmarking
104 across fundamentally different estimator classes [6]. This gap is especially rele-
105 vant in embedded BMS design, where estimator selection must jointly consider
106 nominal accuracy, convergence, robustness under disturbed measurements, and
107 computational constraints.

108 *1.4. Contribution and study objectives*

109 This work addresses the gap through three linked questions: whether clean-
110 data accuracy predicts performance under measurement and signal-integrity
111 disturbances, which structural mechanisms govern degradation and recovery,
112 and how the resulting behavior trades against embedded execution cost. To
113 answer them, four deployment-oriented representatives are executed under one
114 causal protocol on cell-disjoint LFP aging data covering multiple load and SOH
115 states. The campaign combines physically interpretable measurement, timing,
116 availability, quantization, and initialization disturbances with a common recov-
117 ery criterion, cell-level uncertainty analysis, a paired SOH-input ablation, and
118 STM32 measurements of the isolated SOC cores.

Table 1: Estimator-class envelope and scope of the four benchmark representatives. Results apply to the selected implementations in the last column, not automatically to every method listed in the broader class envelope.

Class	Minimum defining mechanism	Broader class can include	Representative used in this study
Direct measurement	Causal integration of measured current using an assumed capacity	Coulombic efficiency, capacity adaptation, OCV or end-point anchoring, plausibility logic, and reset thresholds	Fixed-capacity Coulomb counting with a sustained full-charge CV reset and no voltage-residual correction
Hybrid direct measurement	Integration core augmented by an estimated or learned state/parameter	Learned capacity or efficiency, residual correction, multi-sensor fusion, and adaptive anchoring	Same Coulomb-counting and reset logic as DM, with effective capacity supplied by the shared causal LSTM-SOH estimate
ECM observer	Coulomb-counted SOC propagation plus a voltage model and feedback correction	Higher RC order, hysteresis, thermal states, SOC/SOH/temperature maps, and different nonlinear observers	Second-order RC model with EKF correction, charge/discharge SOC-SOH lookup surfaces, and shared causal SOH input
Data driven	Learned mapping from measured signals or their history to SOC	Feed-forward, convolutional, recurrent, attention-based, physics-informed, raw-signal, or engineered-feature models	Structurally pruned rolling-window GRU using voltage, current, temperature, SOH, Q_c , derivatives, and Δt

The resulting evidence supports comparisons among these four implementations under the declared protocol. It does not establish a universal ranking of direct-measurement, hybrid, observer-based, or data-driven families, whose behavior can change with anchoring logic, parameterization, architecture, features, training data, and deployment semantics.

2. Methodology

2.1. Estimator representatives and class boundaries

Figure 2 summarizes the causal benchmark chain, from measured signals and online feature reconstruction to disturbance injection and global and local evaluation. The four estimators are selected to expose distinct correction mechanisms, not to span every implementation possible within each class. Because class boundaries overlap, each representative is assigned according to its dominant SOC update and correction mechanism. Table 1 separates this defining principle from optional extensions and from the exact implementation evaluated here. There is no meaningful universal “maximum” model within a class: adding learned corrections, higher-order dynamics, richer sensing, or more complex sequence architectures progressively changes both capability and deployment cost.

All SOH-dependent representatives, namely HDM, HECM, and DD, receive the same causal LSTM-SOH trace so that differences arise from how the SOC branch consumes that context. DM is the low-complexity integration reference.

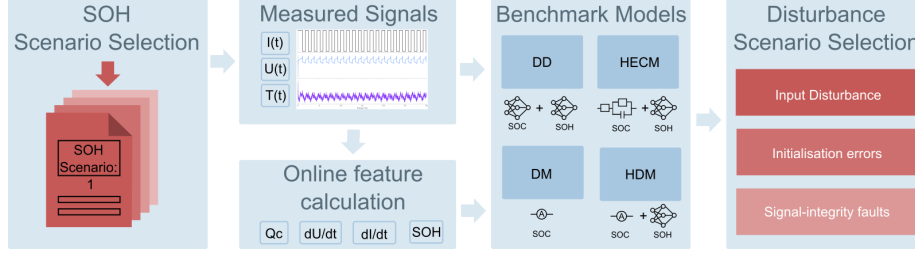


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator implementations, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

139 HDM isolates the effect of learned capacity adaptation without adding volt-
 140 age feedback. HECM adds model-based voltage-residual correction to current-
 141 driven state propagation. DD represents compact temporal learning. The GRU
 142 was selected because SOC and disturbance recovery depend on recent signal his-
 143 tory while its gated single-state structure is lighter than a comparable LSTM.
 144 Because DD also consumes the engineered Q_c feature, it is a data-driven se-
 145 quence estimator with physics-informed inputs rather than a raw-signal-only
 146 neural model.

147 2.2. SOC estimation methods

148 2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c . It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1} converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped

directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$

149 2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

150 This formulation makes the relation between DM and HDM explicit: both use
 151 the same online charge state Q_c , while only the denominator changes through
 152 online SOH support. In both direct-measurement variants, a sustained near-
 153 maximum voltage condition is interpreted as a constant-voltage full-charge event.
 154 If the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible volt-
 155 age margin below the upper threshold U_{max} , for the configured CV duration,
 156 the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$. This re-
 157 flects the practical assumption that a completed CV phase provides a physically
 158 anchored full-charge reference.

159 2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where $C_b = C_0 \widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1} , whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The benchmarked, structurally pruned and fine-tuned model consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as the finite differences $dI/dt(k) = (I_k - I_{k-1})/\Delta t_k$ and $dU/dt(k) = (U_k - U_{k-1})/\Delta t_k$. They provide local transient information that is not captured by the absolute channels alone. The additional feature Δt_k allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is

used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information. The first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC estimators to use the latest SOH context. This cadence matches the slowly varying capacity fade observed in the dataset but cannot represent an abrupt capacity-loss event, which is absent from the measurements and outside the study claims. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and

the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

2.4. Model and target-hardware preparation

The software benchmark uses trained floating-point model parameters and input scalers fixed before holdout evaluation. Neural training, validation, hyperparameter selection, scaler fitting, pruning, and final model selection use only the declared training and validation cells. No holdout cell enters these steps. The imported HECM lookup surfaces were preidentified from separate HPPC experiments and held fixed before this benchmark. No scored trajectory or benchmark result was used to fit or retune them. The compact SOC-GRU has hidden size 67 and the shared SOH-LSTM has hidden size 112. All model configurations, parameter sets, and preprocessing definitions were fixed and documented before evaluation. The evaluation procedure accepts only cells from the declared holdout set unless an analysis is explicitly identified as diagnostic.

The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC implementation cost from the shared SOH service, the board receives the pre-computed causal SOH trace used by the software reference. The SOH-LSTM itself is not executed on the microcontroller. Numerical equivalence and inference latency are measured directly on the board. Execution time is recorded for every valid inference with the processor-integrated hardware cycle counter, and each cell sequence is replayed three times. Flash occupancy is calculated from the compiled firmware regions that must be stored in the microcontroller’s non-volatile program memory.

Runtime RAM is divided into statically allocated variable storage, dynamically allocated memory, and temporary memory used during nested function calls. The static component includes both initialized variables and variables set to zero during system startup. Maximum temporary function-call memory is measured by filling the available stack region with a known pattern be-

246 fore initialization. After a representative 2025-sample replay of cell C09, the
 247 deepest overwritten address identifies the maximum function-call memory used.
 248 Dynamic-memory management is instrumented to record the maximum cumu-
 249 lative allocation reached during execution. The reported peak RAM combines
 250 these three contributions and therefore covers startup, protocol handling, and
 251 estimator execution without counting an unused memory safety reserve config-
 252 ured during compilation as consumed memory. Predictions are compared with
 253 both the software implementation and dataset SOC. The experiment charac-
 254 terizes the isolated SOC implementations rather than end-to-end SOC-SOH
 255 system behavior.

256 *2.5. Robustness benchmark design*

257 The benchmark follows a strict measurement-only principle: only quantities
 258 that a real BMS measures or derives online from measured data are manipu-
 259 lated, namely current, voltage, temperature, sample availability, and sampling
 260 time, whereas artificial parameter mismatch and synthetic hidden-state corrup-
 261 tion are deliberately excluded [6]. Figure 3 groups the benchmark into input
 262 disturbances, initialization errors, and signal-integrity faults, while Table 2 de-
 263 fines the exact disturbance implementations used in the study.

264 The disturbance magnitudes are not intended as one-to-one copies of a single
 265 field specification. Instead, they were selected from three justification classes:
 266 deployment-level sensing limitations, adverse measurement corruption, and de-
 267 liberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark
 268 matrix follows the scenario families shown in Figure 3 and provides the basis
 269 of the main result figures and tables. ADC quantization is part of the same
 270 scenario set and is reported in Figure 16. Overall, the benchmark is designed
 271 as a controlled and reproducible scenario study that exposes estimator-specific
 272 failure modes under a common disturbance protocol rather than emulating the
 273 complete system stack of a production BMS.

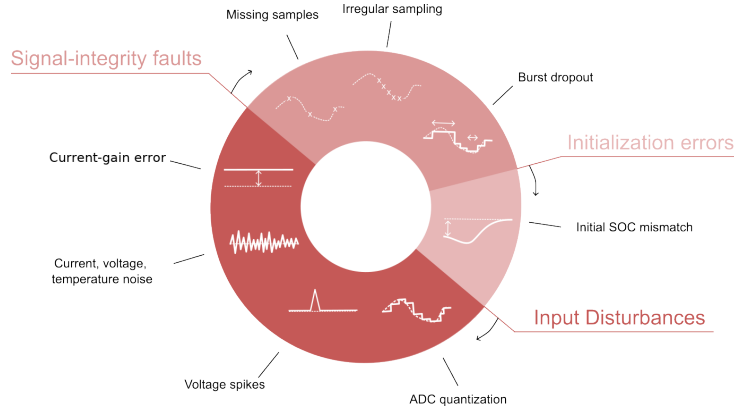


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

Table 2: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings. The cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C	ADC steps, sensor scaling, rounding, and coarse deployment-level discretization [20, 21, 26]
Current gain error	Signed gain sweep: $I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$	Sensor gain error, miscalibration, drift, and upper bound as stress case [21, 22, 23]
Current noise	Gaussian with $\sigma_I = 0.02$ A and 0.10 A	Sensing-chain noise, EMI pickup, and low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian with $\sigma_U = 0.01$ V	Wiring noise, front-end noise, poor filtering, and EMI [4, 20, 24]
Temperature noise	Gaussian with $\sigma_T = 1.0$ °C	Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch [26, 11]

Table 2: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	Benchmark setup	Practical origin / rationale
Voltage offset	Additive $U_{\text{meas}} = U + 0.02 \text{ V}$	Front-end offset, calibration residual, wiring contribution, and adverse sensing stress
Temperature offset	Additive $T_{\text{meas}} = T + 3 \text{ }^\circ\text{C}$	Sensor placement error, calibration residual, and cell-to-sensor thermal mismatch
Initial SOC error	Initial mismatch of -0.10 SOC, with DD perturbed through online Q_c	Restart, storage, missing recent history, and weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen and separate random 2% loss	Packet loss, blocked updates, logger dropouts, and periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1 \text{ s}$, $\pm 0.5 \text{ s}$, and $\pm 0.9 \text{ s}$ around nominal 1 s grid	Scheduling jitter, bus latency, and asynchronous acquisition [6, 5, 25]
Burst dropout	One frozen 3600 s interval selected by maximum absolute net charge, with at least 12 h before and 24 h after the interval	Communication interruption, frozen task, stuck data stream, and measurement-only severe case [6, 5]
Voltage spikes	Single-sample spikes of $\pm 0.20 \text{ V}$ every 1000 samples (1000 s)	EMI, switching disturbance, and corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, recovery after initialization mismatch, and robustness under sustained or transient measurement corruption and signal unavailability. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum

280 error, and the 95th percentile absolute error, because these metrics provide a
 281 compact view of average performance, systematic deviation, and tail behavior
 282 across the selected evaluation windows. Local disturbance performance is an-
 283 alyzed through scenario-specific metrics such as jump counts, output variance,
 284 absolute-error variance, and excess rolling MAE, because several disturbances
 285 do not primarily manifest as window-wide MAE collapse but as short-term
 286 output fluctuations, delayed drift, or slow re-synchronization. Whenever a dis-
 287 turbance mask exists, the analysis further separates pre-disturbance, disturbed,
 288 and post-disturbance behavior in order to quantify disturbance-window error
 289 growth and residual post-disturbance error.

290 ADC quantization, current-gain error, voltage and temperature disturbances,
 291 missing samples, and irregular sampling are interpreted primarily through global
 292 metrics. Burst dropout, voltage spikes, current-noise detail, and initialization
 293 mismatch additionally use dedicated local evidence because their most relevant
 294 effects are episodic. The canonical initialization-recovery endpoint compares
 295 each perturbed estimator trajectory with its cell-, window-, model-, SOH-, and
 296 seed-matched correctly initialized trajectory. Recovery is the first entry into an
 297 absolute trajectory-difference band of 0.02 SOC that lasts continuously for at
 298 least 300 s. Evaluation is right-censored at 24 h. A later departure from the band
 299 is retained as a separate relapse endpoint. The comparison reports recovery-
 300 or-censoring time, censored and relapse fractions, and excess-error area under
 301 the curve. It never substitutes absolute estimator error against dataset SOC
 302 for the paired recovery difference. The requested -0.10 initialization mismatch
 303 is mapped into each estimator’s available online state: the coulomb-counting
 304 SOC initialization for DM and HDM, the EKF SOC state for HECM, and the
 305 capacity-equivalent Q_c offset for DD. Because the DD mapping passes through
 306 the recurrent network rather than directly fixing its output, every realized initial
 307 output difference is reported before recovery is compared. The resulting ranking
 308 compares deployment-accessible initialization interventions, not mathematically
 309 identical latent-state perturbations.

310 The independent statistical unit is the holdout cell, not an individual time

311 sample or selected window. Predeclared SOH windows are first averaged within
 312 each cell and random seed. The six cells then receive equal macro weight. Hier-
 313 archical bootstrap resampling with seeds nested inside cells and 10,000 repeti-
 314 tions provides 95% confidence intervals. Scenario effects are evaluated as paired,
 315 cell-matched differences from baseline, and estimator pairs are compared using
 316 exact two-sided sign-flip tests, raw mean and median differences, paired stan-
 317 dardized effect sizes, and Holm-adjusted p -values. Holm correction is applied
 318 within each reported scenario family. Because only six independent cells are
 319 available, effect magnitudes and confidence intervals are treated as primary ev-
 320 idence and p -values as supporting evidence. The full machine-readable tests
 321 accompany the result tables, while the baseline pair tests are reproduced in the
 322 Appendix.

323 The rolling DD implementation first emits an SOC estimate at source sample
 324 2023 because it requires a 2024-sample causal input window. All global accuracy,
 325 robustness, stratified, confidence-interval, and paired-test calculations therefore
 326 use the common source-sample mask $k \geq 2023$ for every estimator. Each 24-h
 327 window contributes 84,377 matched evaluation points per estimator, while each
 328 48-h dropout window contributes 170,777. This prevents the other estimators
 329 from receiving an additional early interval that DD cannot evaluate.

330 The decision synthesis is explicitly illustrative. Accuracy combines base-
 331 line MAE, RMSE, and 95th-percentile error. Robustness assigns equal weight
 332 to eight declared disturbance families: sensor noise, current-gain error, sensor
 333 offsets, quantization, missing samples, timing jitter, burst dropout, and volt-
 334 age spikes. Non-positive Δ MAE values are treated as zero disturbance penalty.
 335 Levels within a family are averaged before the family scores are combined, so
 336 families with more tested magnitudes do not receive extra weight. Recovery
 337 combines the four outputs from the single canonical paired initialization anal-
 338 ysis: recovery-or-censor time, excess-error area, censored fraction, and relapse
 339 fraction. Alternative equal-scenario and high-severity weighting variants are re-
 340 tained as sensitivity analyses. The normalized synthesis is not treated as an
 341 objective estimator-family ranking.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. Model development and final testing use a fixed split. SOC and SOH training use C01, C03, C05, C11, C17, and C23. Validation uses C07, C19, and C21. The independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. The six test cells span distinct aging, thermal, duration, and load envelopes, as summarized in Figure A.21. LFP’s flat mid-SOC OCV and chemistry-specific aging behavior are integral to the tested problem.

The holdout cells are assigned to descriptive load groups before model results are inspected, using visible separation in the measured 95th-percentile absolute C-rate: C25 and C27 form the low-load group, C09, C13, and C15 the middle-load group, and C29 the high-load case. Because the high-load group contains one cell, its group-by-SOH results are explicitly exploratory and are not interpreted as population-level high-load inference. Within each trajectory, performance is additionally stratified by reference SOH (fresh at $\text{SOH} \geq 0.90$, mid-life at $0.80 \leq \text{SOH} < 0.90$, and aged at $\text{SOH} < 0.80$), instantaneous absolute C-rate (< 0.5 , $0.5\text{--}1.5$, and ≥ 1.5), temperature ($\leq 30^\circ\text{C}$, $30\text{--}35^\circ\text{C}$, and $> 35^\circ\text{C}$), and SOC (< 0.20 , $0.20\text{--}0.80$, and > 0.80). Absent states are reported as not available and are never imputed.

3.2. Reference targets and preprocessing

The reference preprocessing first computes the signed transferred charge as $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$. The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the

371 resulting capacity and SOH columns are interpolated between the dedicated
 372 capacity-test anchors. The reference SOC target used in the benchmark corre-
 373 sponds to the offline Coulomb-counting target $\text{SOC}_{\text{ref}} = 1 + Q_c/C_{\text{ref},0}$, where
 374 the cumulative charge state Q_c is propagated from the signed charge and reset
 375 to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor dur-
 376 ing preprocessing. These reference targets are intentionally treated as offline
 377 evaluation quantities, because they rely on dataset-side processing and anchor
 378 logic that are not fully available to an online estimator during deployment.

379 *3.3. Test trajectory and benchmark scenarios*

380 All estimators are evaluated on every available holdout-cell/SOH combina-
 381 tion under the same disturbance protocol. One representative 24-h window is
 382 frozen for each available fresh, mid-life, and aged state, yielding 16 windows.
 383 C27 contributes only a fresh window because its measured SOH does not fall
 384 below 0.90. Windows start at a measured full-charge anchor ($\text{SOC} \geq 0.98$,
 385 $U \geq 3.58 \text{ V}$) and are selected as measured-feature medoids without using es-
 386 timator outputs or errors. The one-hour dropout uses 48 h from the same
 387 anchor to retain approximately 24 h of post-gap observation. The shared SOH
 388 LSTM receives 192 h of undisturbed causal context before every scored window,
 389 matching its trained sequence length. Figure A.23 summarizes this protocol.

390 The 19 predeclared cases include the nominal baseline, two current-noise
 391 levels, voltage and temperature noise, three current-gain levels, voltage and
 392 temperature offsets, ADC quantization, initial SOC mismatch, periodic and
 393 random sample loss, three timing-jitter levels, a one-hour burst dropout, and
 394 randomized-sign voltage spikes. Gaussian sensor-noise cases use 10 determinis-
 395 tic seeds. Random sample loss, jitter, and spike cases use 5 seeds. Deterministic
 396 cases run once. Figure A.22 maps each case to the manipulated measurement
 397 or state and marks the selected paired reference-SOH ablations. Input distur-
 398 bances are applied directly to measured channels, initialization errors only where
 399 structurally equivalent, and signal-integrity scenarios to sample availability or
 400 timing while preserving causal estimator execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c , EFC, dU/dt , dI/dt , and hourly SOH aggregates, so no hidden dataset-side feature leaks into the estimator input. The SOH LSTM forwards hidden and cell states across hourly updates after its 192-h initialization context. The primary SOC-GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. The dropout start is selected from measured current only as the eligible one-hour interval with the largest absolute integrated net charge. Selection enforces at least 12 h of pre-gap observations and 24 h of post-gap observations and never uses reference SOC or estimator output. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online Q_c because the network exposes no explicit SOC state. The environment enforces identical capacity assumptions, disturbed streams, targets, and metrics across all estimators.

4. Results

4.1. Baseline performance

Figure 4 summarizes nominal performance across the 16 frozen windows on the common source-sample interval. Equal weighting of the six holdout cells yields a different ranking from the previous single-trajectory analysis: DD has the lowest baseline MAE at 0.0258 (95% CI 0.0189–0.0325), followed by HECM at 0.0337 (0.0246–0.0434), HDM at 0.0412 (0.0301–0.0515), and DM at 0.0690 (0.0499–0.0823). The RMSE ranking is identical, with values of 0.0300, 0.0388,

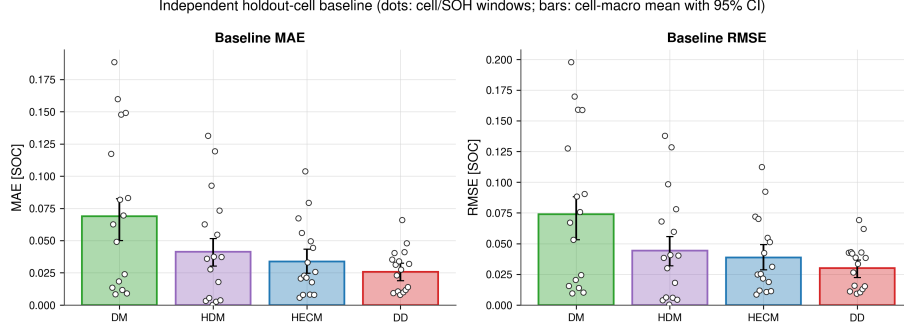


Figure 4: Baseline performance across six independent holdout cells. Dots show cell/SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

0.0442, and 0.0740, respectively. Cell-level points reveal substantial between-cell variation, particularly for DM, so the bars must be interpreted as cell-macro summaries rather than performance on a homogeneous trajectory. Exact paired sign-flip tests support the magnitude and direction of the principal contrasts, but none of the six baseline pair tests remains below 0.05 after Holm correction because only six independent cells are available. Confidence intervals and paired effect sizes are therefore the primary evidence, as reported in the Appendix.

4.2. Current-gain sensitivity

Figure 5 summarizes the adverse response to a signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting worst-case degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0108 for DM, 0.0101 for HDM, 0.0060 for HECM, and 0.0038 for DD.

The continuous C29 lifecycle analysis in Figure 6 separates three mechanisms that independent 24-h windows cannot resolve. The gain-error contribu-

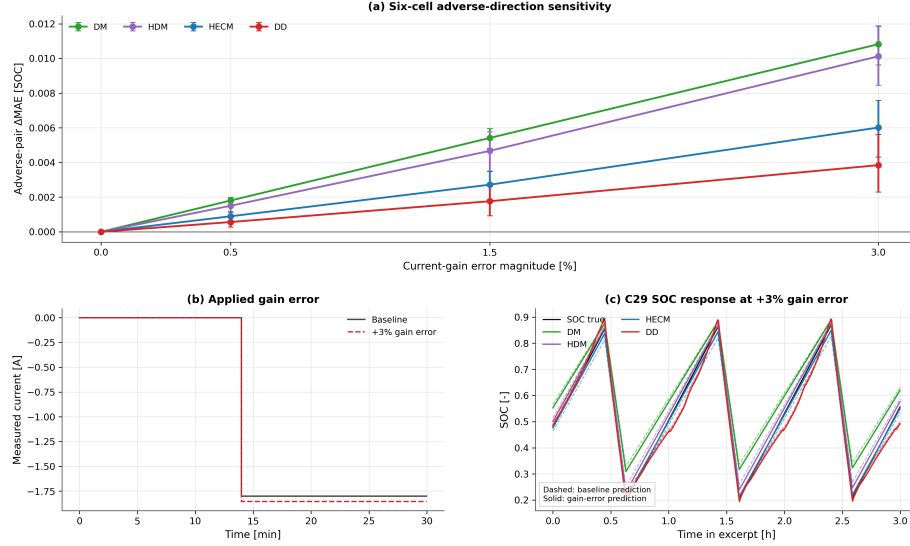


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm b$ pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) C29 SOC trajectories over three charge–discharge intervals. The adverse-pair summary prevents signed compensation of pre-existing baseline bias from being misread as improved robustness.

tion varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty generally grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across the full replay, DD retains the smallest adverse penalty, whereas DM and HDM remain most exposed to accumulated current error.

4.3. Noise sensitivity

Repeated sensor noise is weaker than persistent current gain error, but the response depends on channel and estimator structure. At $\sigma_I = 0.10$ A, HECM has the largest mean macro penalty ($\Delta\text{MAE} = 0.0038$), although its interval spans zero, whereas DM changes by -0.0009 and HDM and DD remain near zero. Small negative changes are paired finite-window effects, not evidence that

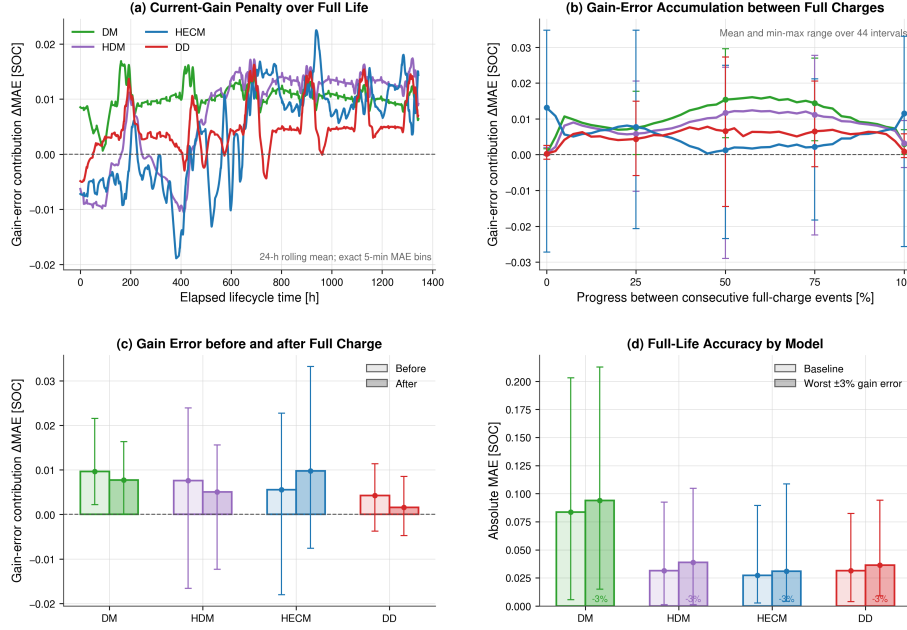


Figure 6: Current-gain-error accumulation over the continuous C29 life replay. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

noise improves an estimator in general. Voltage noise ($\sigma_U = 0.01$ V) produces a 0.0014 penalty for DM and 0.0013 for HDM, while HECM and DD remain below 0.0003. Temperature noise is neutral for DM and small for the SOH-aware models. Figure 7 shows the local transmission of current noise, while Figure 8 reports repeated six-cell effects with hierarchical uncertainty.

The offset cases expose a different pattern from zero-mean noise. A $+3^\circ\text{C}$ temperature offset increases DD MAE by 0.0074 (95% CI 0.0027–0.0125), while the corresponding changes for DM, HDM, and HECM are 0.0000, 0.0002, and 0.0002. A $+0.02$ V voltage offset increases DD by 0.0038 and HDM by 0.0005, leaves DM effectively unchanged, and reduces HECM MAE by 0.0040 in these

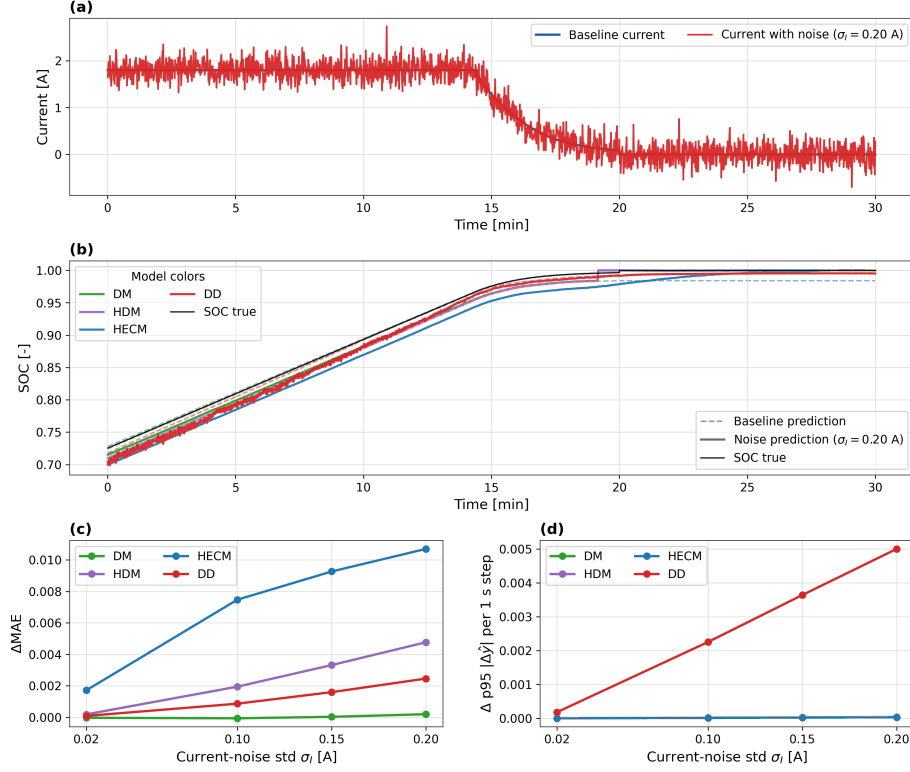


Figure 7: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

finite windows. The latter signed compensation is not interpreted as a generally beneficial sensor fault. These offset results are included in the cross-scenario heatmap and in the family-balanced robustness synthesis.

4.4. Initialization error and recovery

Initialization performance uses the paired criterion defined above. The same requested -0.10 SOC-equivalent mismatch is mapped to the explicit DM/HDM Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$ for DD. Figure 9 reports the realized trajectory difference rather than assuming that these internal mappings create identical outputs. All four paired trajectories use source samples 2023 onward, while time remains referenced to the

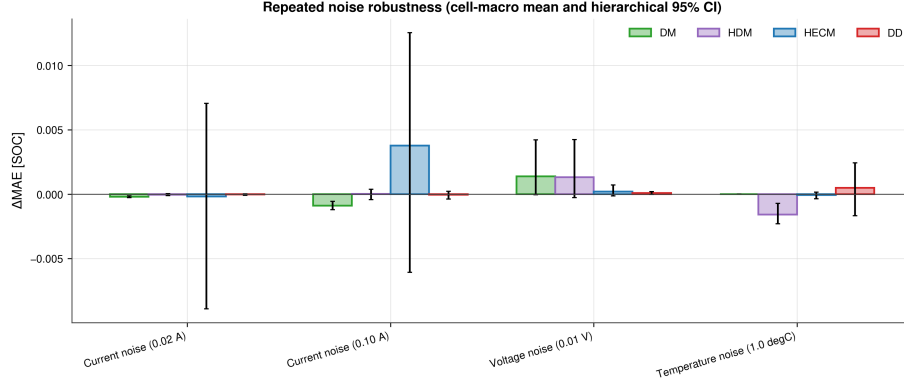


Figure 8: Repeated noise robustness across six holdout cells. Bars show cell-macro Δ MAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

481 intervention at sample 0. HECM reaches the first qualifying 300-s band in
 482 1.12 h on average and DD in 1.29 h. HDM and DM require 17.21 and 22.28 h,
 483 respectively. DD’s cell-macro relapse fraction after a first qualifying hold is
 484 0.78, compared with 0.17 for HECM. DM and HDM are right-censored in cell-
 485 macro fractions of 0.89 and 0.67. HECM also has the smallest excess-error
 486 area at 0.078 SOC h, compared with 0.116 SOC h for DD. Considering time,
 487 excess-error area, censoring, and relapse together, HECM provides the strongest
 488 recovery profile for this intervention. Integration-based states generally require
 489 a later physical anchor.

490 4.5. Missing samples and signal integrity

491 Figure 10 compares the six-cell effects of periodic and random sample loss
 492 with those of timing jitter. Sample loss is substantially more consequential
 493 than timing jitter. Periodic freezing of every 50th sample increases MAE by
 494 0.0072, 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively.
 495 Random 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and
 496 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately
 497 6×10^{-4} SOC of baseline for every model, with confidence intervals generally

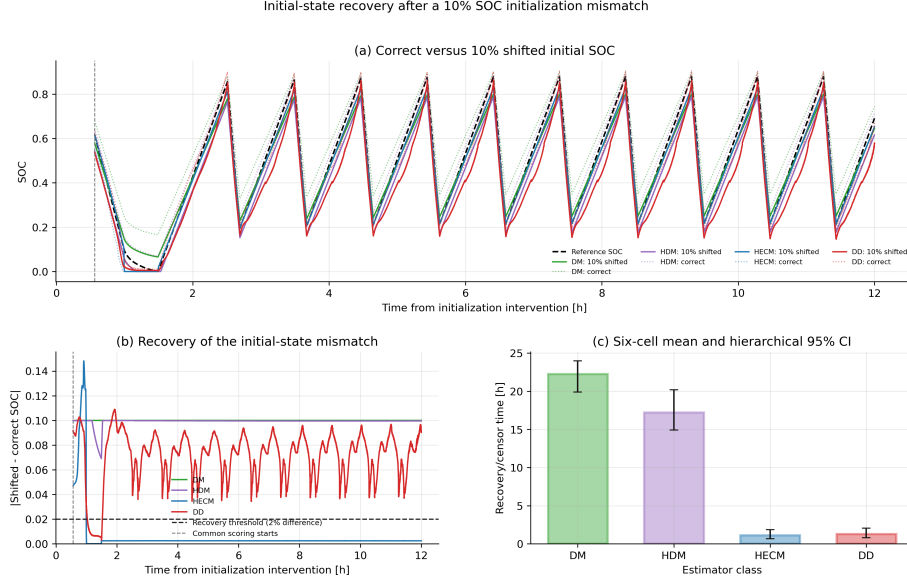


Figure 9: Recovery after a 10% SOC-equivalent initialization mismatch. (a) Representative paired trajectories on the common source-sample interval. (b) Difference between shifted and correctly initialized outputs with the common 2% threshold. Time is measured from the intervention at sample 0, and the vertical marker denotes the first commonly scored sample. The DD example illustrates why later relapse is recorded separately. (c) First 300-s recovery-or-censor time across the six holdout cells with hierarchical 95% confidence intervals.

spanning zero. DD therefore has the smallest sample-loss penalty, while DM and HDM retain their structural dependence on uninterrupted charge integration.

The one-hour burst dropout is a distinct availability problem. Against the newly generated duration-matched 48-h baseline, its macro ΔMAE is largest for HECM (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and DM (0.0050, -0.0029 –0.0138). DD changes by -0.0093 (-0.0288 –0.0099). The signed DD value indicates finite-window compensation between an already imperfect baseline trajectory and the frozen-input trajectory, not that missing measurements are beneficial. The interval is deliberately placed where the absolute unobserved net charge is largest among starts that retain the required pre- and post-gap context. Figure 11 compares each disturbed C29 output with its duration-matched no-dropout trajectory and reports the six-cell

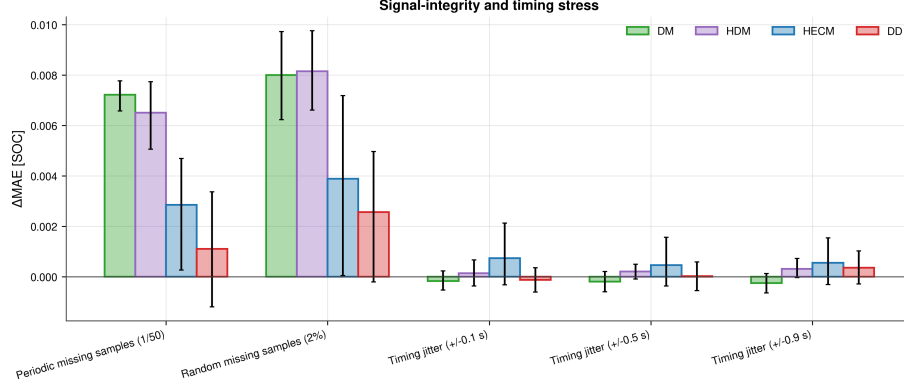


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

510 global effect. The paired local deviation has model-specific origins: DM/HDM
 511 retain the unobserved charge deficit until a physical anchor, HECM corrects
 512 through weak LFP voltage observability, and DD changes as its rolling input
 513 context is replaced by valid samples. No six-cell dropout-recovery time is used
 514 in the recovery score because recovery is defined only by the canonical paired
 515 initialization experiment.

516 4.6. Spike sensitivity

517 Single voltage spikes demonstrate why local and global metrics must be re-
 518 ported together. Full-window ΔMAE remains below 4×10^{-4} SOC for every
 519 estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis,
 520 however, identifies a reproducible DD transient: its mean spike-sample penalty
 521 across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct
 522 penalties for the other representatives. Figures 12 and 13 further show that the
 523 DD response varies across cells and SOH states and then decays after the event.
 524 This pattern is consistent with a one-sample voltage impulse entering both the
 525 absolute voltage and dU/dt channels and perturbing the rolling recurrent pre-
 526 diction. It is a mechanism-supported interpretation, not proof of a unique causal
 527 input channel because no channel-removal ablation was performed.

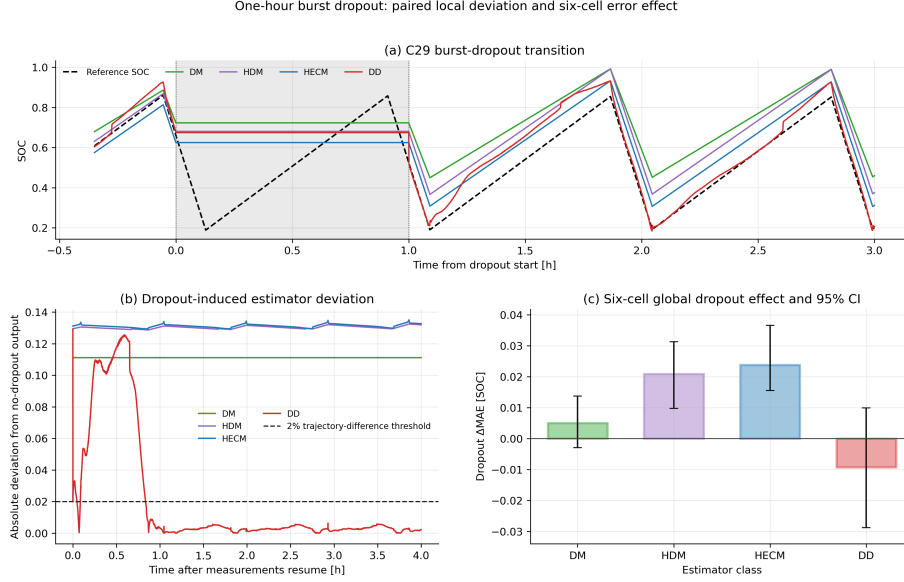


Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global Δ MAE with hierarchical 95% confidence intervals.

4.7. Shared-SOH ablation

The primary comparison intentionally supplies one common causal LSTM-SOH trace to HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset reference SOH to quantify this shared-service confounder. Reference SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, but increases DD MAE by 0.0049 on average. The same direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH still improves HDM and HECM but worsens DD by 0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by only 0.0016 for HDM and less than 10^{-4} for HECM and DD, indicating that the incremental temperature-noise response is not dominated by the shared SOH service. This does not imply that a less accurate SOH estimate is intrinsically beneficial. Instead, the models consume SOH differently and were

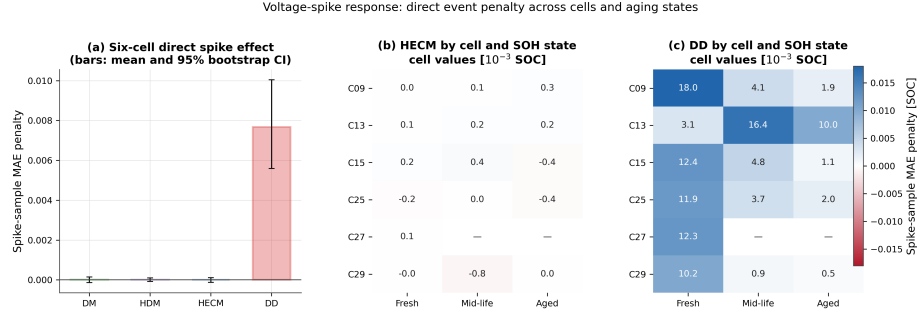


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window Δ MAE.

541 calibrated with different feature interactions. The LSTM-SOH primary results
 542 therefore represent deployable pipeline performance, while reference-SOH runs
 543 are explanatory ablations rather than replacement rankings.

544 4.8. Robustness synthesis across representatives

545 Figure 14 condenses the complete measurement-only campaign and tests
 546 whether the nominal-accuracy ranking alone describes degraded-input behav-
 547 ior. It does not. Persistent gain error and missing information create large global
 548 penalties. Current-gain sensitivity increases with magnitude for every represen-
 549 tative, periodic and random sample loss particularly affect DM and HDM, and
 550 the one-hour dropout produces the largest penalty for HECM. Timing jitter re-
 551 mains nearly neutral. Voltage spikes illustrate the opposite limitation of global
 552 averaging because their full-window MAE is small even though event alignment
 553 exposes a distinct DD transient. The eight-family aggregate shown later is only
 554 one declared weighting of these heterogeneous effects. Its ranking is compared
 555 with alternative weightings rather than interpreted as an objective robustness
 556 winner. The central result is therefore a multi-dimensional and scenario-specific
 557 comparison rather than one universal ranking.

558 Under the family-balanced definition, the illustrative robustness scores are

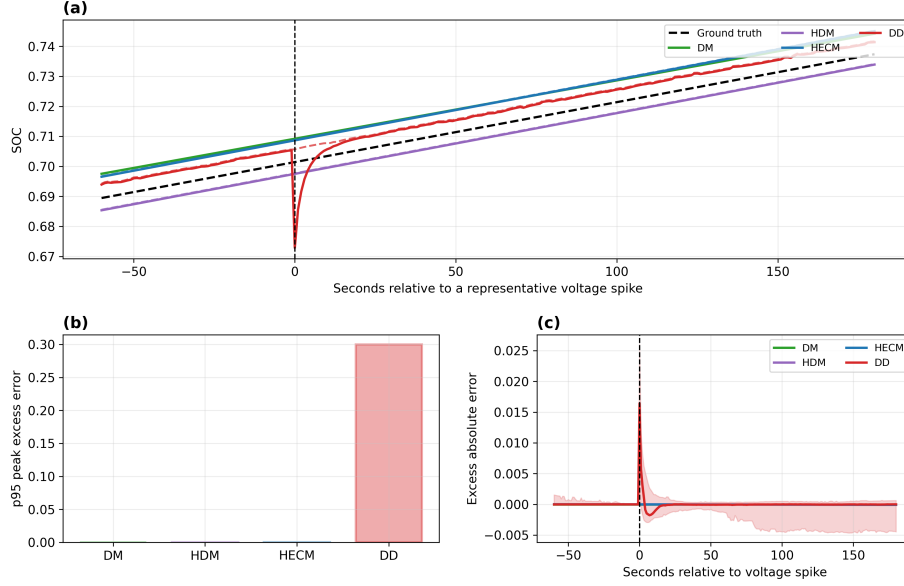


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing reference SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

559 0.562 for DM, 0.468 for HDM, 0.428 for HECM, and 0.842 for DD. Equal weight-
 560 ing per declared scenario yields 0.574, 0.504, 0.560, and 0.781, while the highest-
 561 level-plus-offset variant yields 0.615, 0.512, 0.576, and 0.665. DD remains the
 562 leading tested representative under these three declared aggregations, but its
 563 score and the ordering of HDM and HECM change materially. The result
 564 therefore supports DD for the tested disturbance mix without establishing an
 565 aggregation-independent ranking of estimator classes. The individual scenario
 566 results remain the primary robustness evidence.

567 4.9. ADC quantization

568 Figure 16 compares representative raw and quantized current and voltage
 569 signals and summarizes the resulting six-cell ΔMAE . The tested ADC steps
 570 ($\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C) preserve the local signal shape

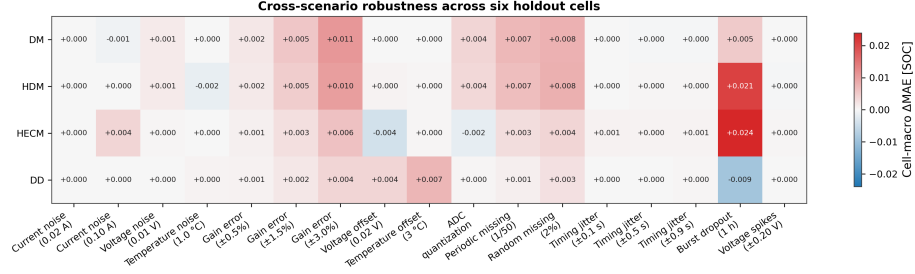


Figure 14: Cell-macro Δ MAE across the complete six-cell measurement-only campaign. Positive values denote degradation relative to each model’s paired baseline. Small negative values reflect finite-window signed effects and are not interpreted as beneficial disturbances.

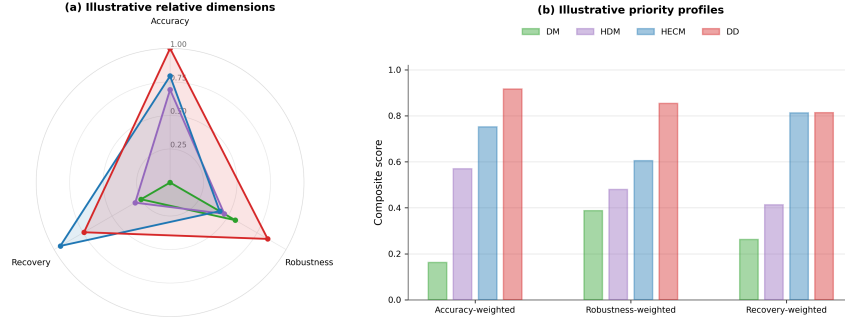


Figure 15: Decision-oriented synthesis for the four tested representatives. (a) Relative min-max-normalized accuracy, family-balanced robustness across all eight declared disturbance families, and canonical paired recovery. (b) Composite scores under three illustrative 60/20/20 priority profiles. Scores summarize the present benchmark only, depend on the declared weighting, and do not replace scenario-level evidence.

571 but affect the tested representatives differently. DM and HDM increase by
572 0.0038 and 0.0035 MAE, respectively. HECM changes by -0.0018 and DD by
573 $+0.0002$. All four confidence intervals overlap zero. Thus, quantization at these
574 deliberately coarse steps is secondary to current gain error, sample loss, and
575 burst dropout in this dataset, but it is not negligible for the integration-based
576 models.

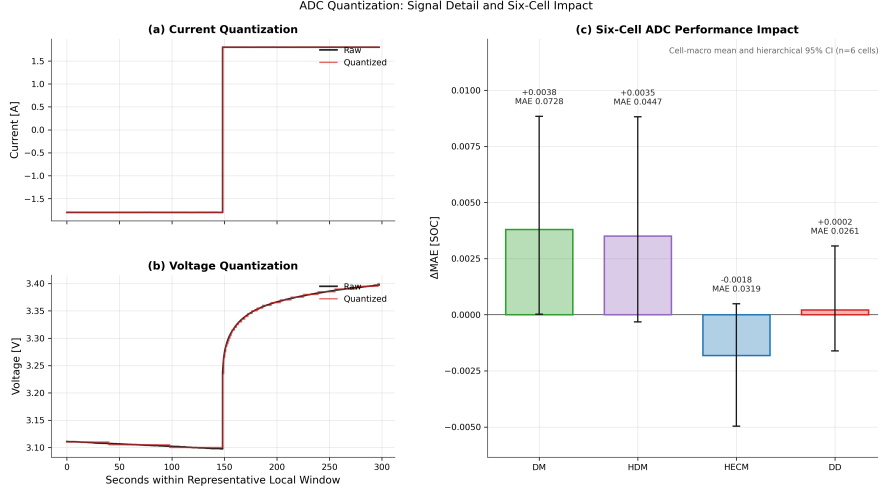


Figure 16: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell Δ MAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

4.10. Embedded performance benchmark

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core and reaches 3.41×10^{-6} percentage points for DD (Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μ s), HECM requires 23.9 μ s, and continuous-state DD requires 424.9 μ s (Figure 18). At the nominal 1-Hz update rate, these medians occupy less than 0.043% of one sampling period, although this isolated deadline share is not a measurement of total BMS CPU load. The DD value in this cross-model comparison is the continuous-state deployment variant. Exact 2024-sample rolling-window semantics are evaluated separately because recomputing the full sequence at every sample is much slower.

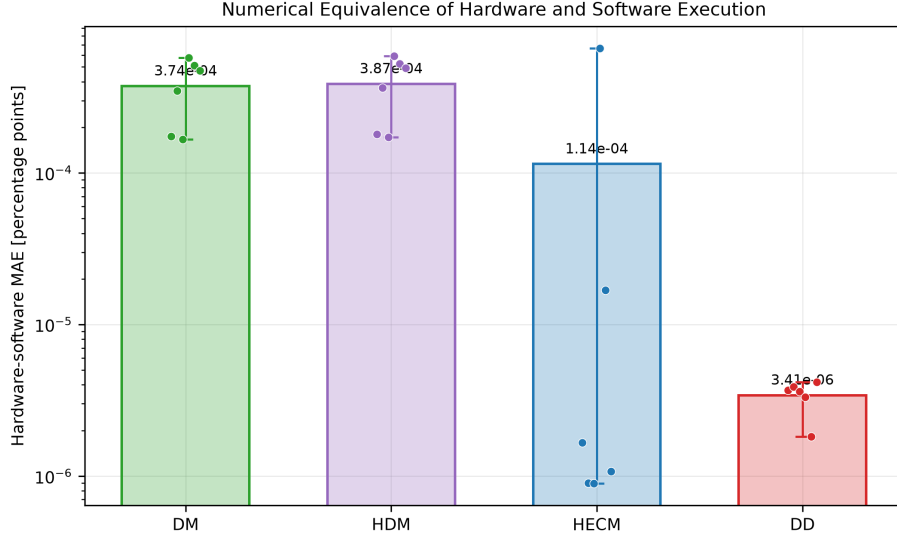


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

For the cross-model embedded performance comparison, DD denotes the continuous-state deployment variant used in the latency analysis. The compiled DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash, respectively. The corresponding measured peak runtime RAM is 2.4, 2.4, 2.5, and 4.1 KiB (Figure 19). In bytes, DM and HDM each combine 872 B of statically allocated variable storage with a 320 B maximum dynamic-memory allocation and 1240 B of maximum call-stack use, yielding 2432 B. HECM reaches 2528 B because its static variable allocation increases to 968 B. Continuous-state DD combines 2588 B of static variable storage with 320 B of maximum dynamic-memory allocation and 1248 B of maximum call-stack use, yielding 4156 B. HECM flash occupancy is dominated by its parameter surfaces, whereas the DD parameter storage accounts for most of its flash demand. These firmware-level values do not include the shared SOH-LSTM service.

On-Device Inference-Latency Distributions Across Six Cells

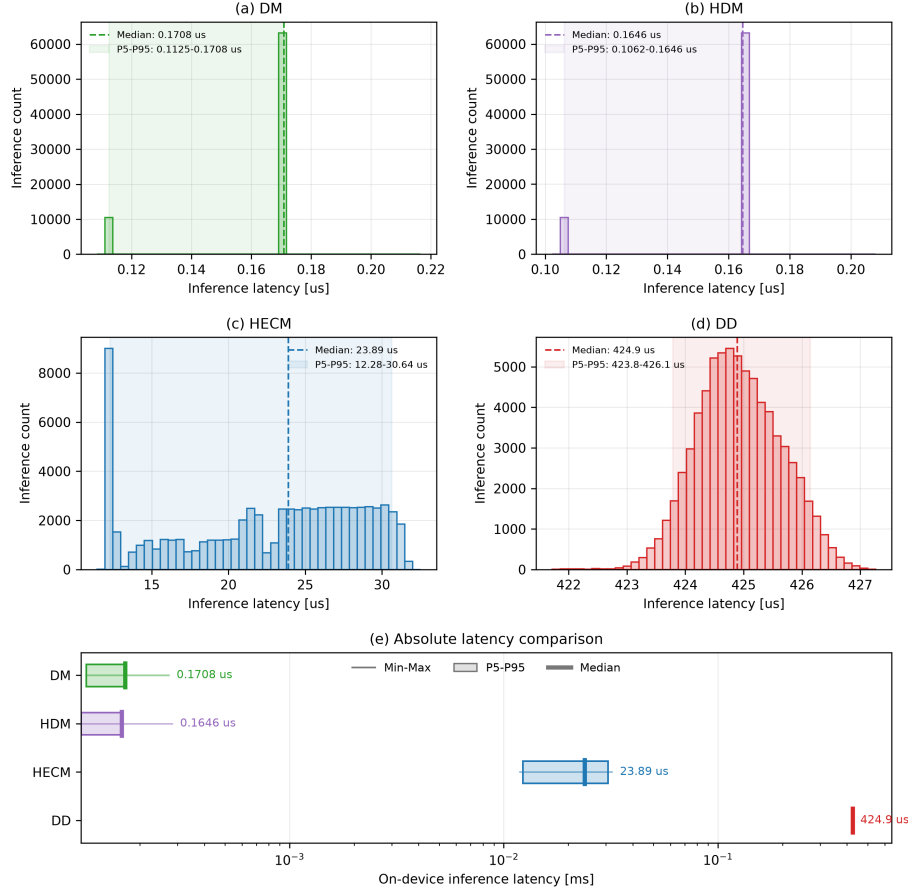


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

Figure 20 makes the DD deployment trade-off explicit, while Figure A.24 provides the corresponding latency distributions over all six hardware cell sequences. Exact rolling-window inference preserves the trained software semantics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 67.3 KiB peak runtime RAM. Continuous-state inference re-

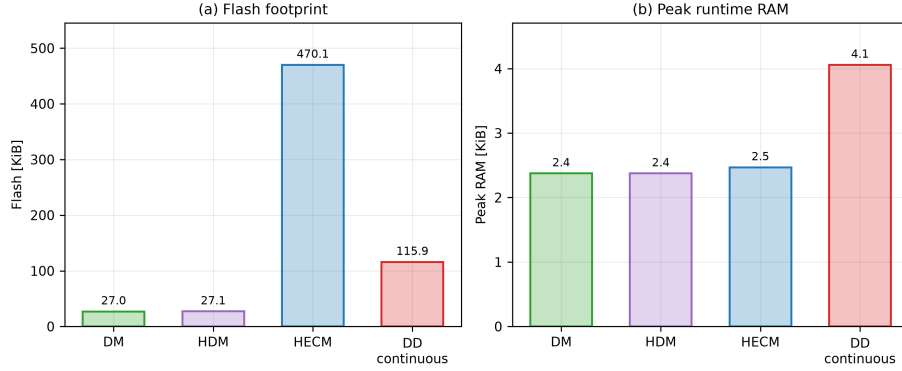


Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Measured peak runtime RAM, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

612 duces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retain-
613 ing similar error on the six nominal sequences. Periodically resetting that state
614 has a comparable 4.1 KiB peak but creates much larger transient errors, with a
615 maximum SOC error near 28 percentage points. Continuous state is therefore
616 the practical hardware candidate, but it remains an inference-semantics abla-
617 tion rather than a drop-in replacement for the primary rolling-window software
618 benchmark.

619 5. Discussion

620 5.1. Structural interpretation of the representatives

621 Because one concrete implementation represents each broad estimator class,
622 the results support mechanism-level interpretation but not family-wide perfor-
623 mance claims. DM behaves as expected for a current-integration representative
624 because it has the largest nominal error and is sensitive to persistent gain error
625 and missing samples when no voltage-feedback correction is available. Its practi-
626 cal full-charge anchor limits unbounded drift, but the lifecycle experiment shows

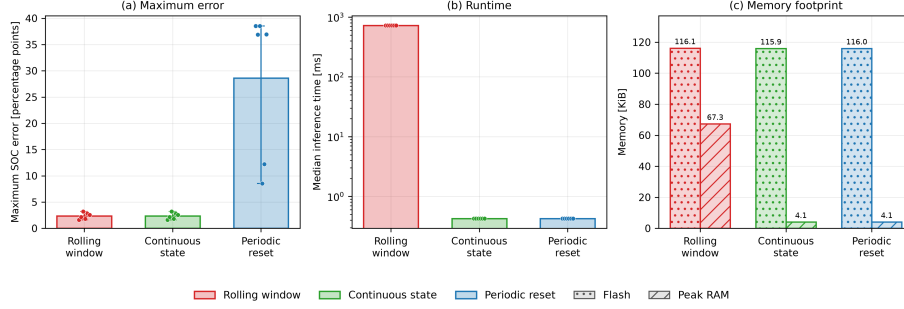


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

627 that the reset does not erase the accumulated gain-error contribution uniformly.
628 HDM adapts the capacity denominator through learned SOH and can reduce
629 capacity mismatch, yet it retains the same charge-integration core. Its simi-
630 lar response to current-gain error and missing charge information is therefore
631 structural to this implementation rather than removed by SOH adaptation.

632 HECM combines current-driven state propagation with a voltage residual
633 and recovers the imposed initial mismatch fastest. This correction capability
634 does not make it universally robust: high current noise and burst dropout pro-
635 duce the largest HECM penalties because both state propagation and observer
636 correction are affected when the causal input stream is corrupted or frozen.
637 The primary campaign does not perturb the fixed HPPC-derived parameter
638 surfaces, so the HECM current-gain result is conditional on this lookup-table
639 quality rather than evidence of equal robustness under parameter misidentifi-
640 cation. Its table dependence and flash footprint must therefore be balanced
641 against its rapid initialization correction.

642 DD has the lowest nominal cell-macro error and the best adverse current-gain
643 score in this comparison. It is also comparatively resilient to isolated sample
644 loss. Conversely, event-aligned voltage spikes expose a recurrent transient that

645 full-window MAE hides. This vulnerability is consistent with the selected volt-
646 age and derivative features and rolling recurrent context. It is specific to the
647 frozen GRU, feature set, training data, and inference semantics, not a universal
648 limitation of recurrent or data-driven SOC estimation.

649 5.2. *Why nominal accuracy is insufficient*

650 Nominal MAE alone fails to reveal the conditions under which the clean-data
651 ranking changes or becomes incomplete. It does not expose the DD voltage-spike
652 transient, the persistence of HECM after a first recovery hold, the shared-SOH
653 feature coupling, or the orders-of-magnitude difference in execution cost. It
654 also cannot resolve how alternative weights across heterogeneous disturbance
655 families change a normalized robustness score. No single scalar captures these
656 deployment trade-offs. Selection should therefore begin with the relevant dis-
657 turbance and recovery requirements and then consider latency, memory, and
658 state-service architecture rather than treating baseline accuracy as a sufficient
659 proxy.

660 5.3. *Implications for BMS deployment*

661 Among the tested representatives, DD is the nominal-accuracy leader and
662 remains a strong candidate when its recurrent implementation cost and voltage-
663 spike response are acceptable. HECM is preferable when persistent correc-
664 tion after state mismatch dominates and a larger flash image is available. DM
665 and HDM remain attractive when deterministic sub-microsecond execution and
666 small memory dominate, but persistent current calibration, reliable sampling,
667 and full-charge anchoring then become stronger system-level requirements. These
668 are implementation-level recommendations, not claims that one estimator fam-
669 ily is intrinsically superior.

670 The STM32 results narrow the gap between algorithmic and deployment
671 claims. All four isolated SOC cores meet the one-second deadline with the
672 tested implementation, but their margins differ by more than six orders of mag-
673 nitude between DM/HDM and exact rolling-window DD. Continuous-state DD

removes most of that cost, at the price of changing inference semantics. The hardware test includes firmware-level measurements of maximum call-stack and dynamic-memory use but does not include the shared SOH-LSTM, energy, pack communication, or concurrent BMS tasks. It therefore supports SOC-core feasibility, not complete BMS schedulability. Table 3 summarizes the evidence without hiding these boundaries.

Figure 15 complements the recommendation map with relative scores, not additional measurements. Lower-is-better inputs are min-max normalized within the four tested implementations, and non-positive disturbance changes contribute zero robustness penalty. Accuracy combines baseline MAE, RMSE, and 95th-percentile error. Robustness gives equal weight to the eight declared disturbance families after averaging levels within each family. Recovery uses only the canonical paired initialization endpoints. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. A separate sensitivity analysis compares this family-balanced definition with equal weighting per scenario and a high-severity variant. The resulting scores are illustrative because their ordering can depend on the declared aggregation rule. They neither remove scenario-specific weaknesses nor support extrapolation from these representatives to their complete model families.

5.4. Limitations and future extensions

Only one frozen implementation represents each broad estimator family. Different reset logic, ECM order and parameterization, neural architecture, input features, training data, or inference semantics can change the ranking. The study is further limited to six holdout cells from one controlled LFP dataset and therefore cannot establish transfer to NMC/NCA chemistry, other form factors, pack imbalance, other thermal environments, or field duty cycles. The load groups are descriptive strata with $n = 2$ low-load, $n = 3$ middle-load, and one exploratory high-load cell. They are not randomized treatments. C27 contains no mid-life or aged window, and missing states are not imputed. With six independent cells, confidence intervals and effect magnitudes are more informa-

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest tested representative when one deployment priority dominates the design decision.

Primary deployment priority	Selected representative	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error	DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Most persistent recovery from initialization mismatch	HECM	Low first-hold time without censoring and substantially fewer later relapses than DD	Highest flash footprint and larger dropout penalty
Lowest adverse current-gain penalty	DD	Δ MAE 0.0038 at 3%, below HECM, HDM, and DM	Temperature-offset and local voltage-spike sensitivity remain visible
Lowest periodic/random sample-loss penalty	DD	Δ MAE 0.0011/0.0026	The duration-matched dropout effect is non-positive but highly variable across cells
Lowest deterministic compute cost	DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment	DD continuous state	Approximately 0.425 ms median and 4.1 KiB peak runtime RAM	Different recurrent semantics from the primary rolling-window benchmark

704 tive than dichotomized significance decisions.

705 Only measurement, timing, availability, and initialization channels are ma-
706 nipulated. Cell/pack parameter mismatch, ECM aging drift, hysteresis er-
707 ror, thermal gradients, sensor cross-sensitivity, balancing and contactor faults,
708 communication-protocol failures, numerical faults, and deadline interference
709 remain outside the campaign. DD initialization is implemented through an
710 SOC-equivalent Q_c offset rather than direct hidden-state corruption. Abrupt
711 capacity-loss events are absent, so the hourly SOH service is evaluated only for
712 gradual aging. The voltage-spike mechanism is supported by event alignment
713 but not proven by an input-channel ablation. HECM results remain conditional
714 on the imported HPPC lookup tables, which were fixed rather than reidentified
715 for each holdout cell.

716 The hardware experiment isolates SOC cores and passes SOH as an exter-
717 nal causal trace. Its memory profile covers the complete benchmark firmware
718 but not an end-to-end SOH service or concurrent BMS tasks. Energy, total
719 scheduled CPU utilization, and pack-level closed-loop behavior also remain out-

side the experiment. Future work should therefore extend memory and energy profiling to the combined SOC–SOH service, add scheduling tests under current BMS workloads, and investigate clustered faults, hidden-state behavior, additional chemistries, and further duty cycles.

6. Conclusion

This study benchmarked the robustness of representatives from four SOC-estimation classes under disturbed sensing and embedded constraints. Nominal accuracy alone is insufficient for estimator selection. DD provides the lowest nominal error, but its local voltage-spike response, initialization relapses, and rolling-window cost are invisible in a clean-data MAE ranking. HECM provides the most persistent recovery from the tested initial mismatch, but is less tolerant of current noise and prolonged dropout and requires the largest flash image. DM and HDM execute with minimal latency and memory, yet remain most dependent on accurate, continuous current integration and physical re-anchoring. Aggregate robustness scores are weighting-dependent and therefore remain illustrative rather than defining a universal winner.

The practical outcome is therefore a selection framework rather than a universal class ranking. Accuracy characterizes nominal prediction, robustness identifies disturbance-specific degradation, recovery describes re-synchronization, and hardware measurements determine whether the trained execution semantics remain practical. The STM32 tests confirm feasibility of all isolated SOC cores while exposing substantial implementation trade-offs. These conclusions apply to the selected Coulomb-counting, SOH-adaptive, ECM–EKF, and GRU representatives on the tested LFP aging data. Broader claims require additional implementations, chemistries, operating profiles, and complete SOC–SOH hardware scheduling.

Appendix A. Holdout Coverage and Frozen Protocol

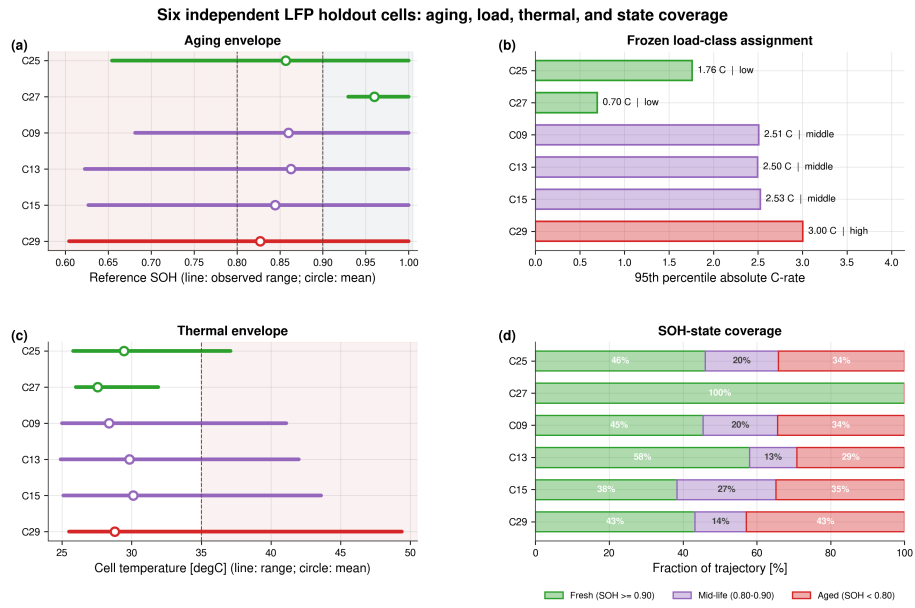


Figure A.21: Quantitative coverage of the six independent holdout cells. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. High-load evidence consists only of C29 and is exploratory.

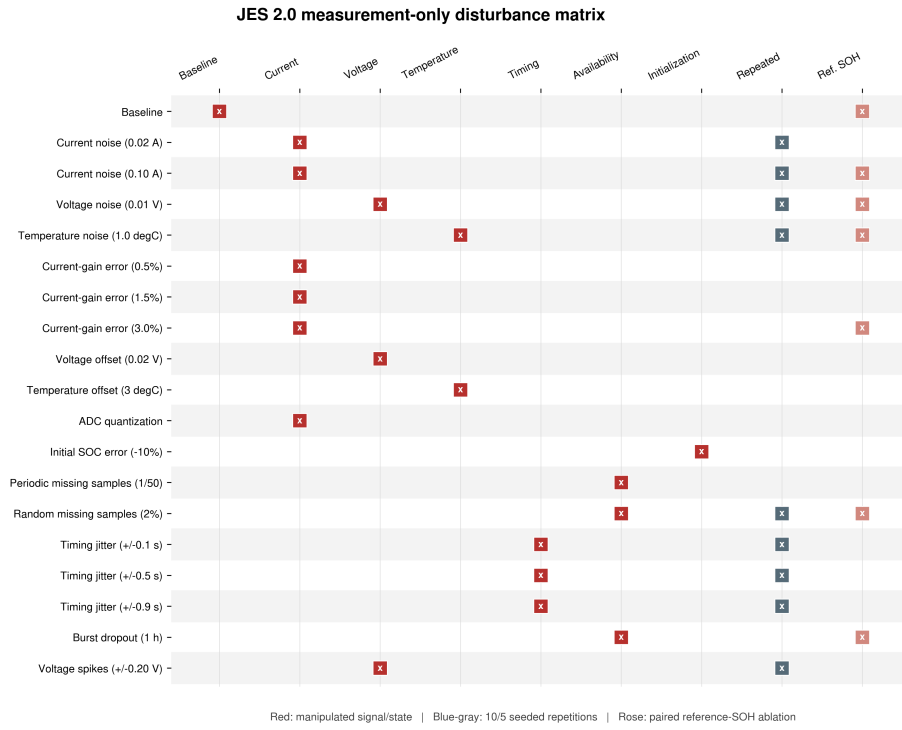


Figure A.22: Complete 19-case JES2 matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations.

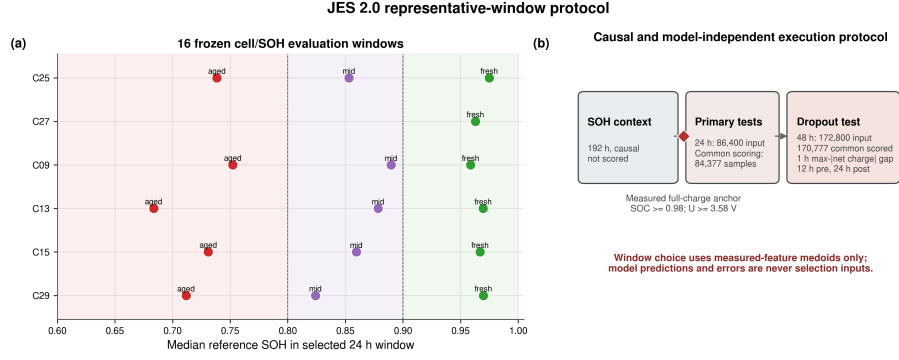


Figure A.23: Frozen representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids only. Every estimator is scored from source sample 2023, yielding 84,377 matched points per 24-h window after a 192-h causal SOH context. The dropout case uses 48 h, places the gap at the eligible one-hour interval with maximum absolute net charge, and contributes 170,777 matched points.

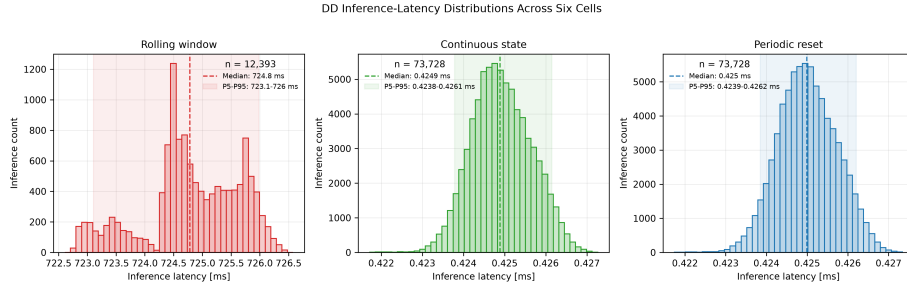


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

747 **Appendix B. Supplementary Numerical Results**

Table B.4: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0690 [0.0499, 0.0823]	0.0740 [0.0532, 0.0881]	0.1088 [0.0768, 0.1307]
HDM	0.0412 [0.0301, 0.0515]	0.0442 [0.0319, 0.0557]	0.0649 [0.0449, 0.0846]
HECM	0.0337 [0.0246, 0.0434]	0.0388 [0.0286, 0.0493]	0.0644 [0.0468, 0.0821]
DD	0.0258 [0.0189, 0.0325]	0.0300 [0.0223, 0.0366]	0.0508 [0.0375, 0.0619]

Table B.5: Exact paired baseline-MAE comparisons across the six holdout cells. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

Estimator A	Estimator B	Mean difference [SOC]	95% CI	d_z	$p_{\text{exact}} / p_{\text{Holm}}$
DM	HDM	-0.0278	[-0.0381, -0.0171]	-1.93	0.0313 / 0.1875
DM	HECM	-0.0352	[-0.0470, -0.0224]	-2.00	0.0313 / 0.1875
DM	DD	-0.0431	[-0.0558, -0.0286]	-2.29	0.0313 / 0.1875
HDM	HECM	-0.0075	[-0.0162, 0.0015]	-0.61	0.1563 / 0.1875
HDM	DD	-0.0154	[-0.0243, -0.0084]	-1.39	0.0313 / 0.1875
HECM	DD	-0.0079	[-0.0176, -0.0005]	-0.68	0.0938 / 0.1875

Table B.6: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from each matched $\pm 3\%$ pair.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10$ A	-0.0009	+0.0000	+0.0038	-0.0000
Current gain, adverse 3%	+0.0108	+0.0101	+0.0060	+0.0038
Voltage offset, +0.02 V	+0.0000	+0.0005	-0.0040	+0.0038
Temperature offset, +3 °C	+0.0000	+0.0002	+0.0002	+0.0074
ADC quantization	+0.0038	+0.0035	-0.0018	+0.0002
Random missing samples, 2%	+0.0080	+0.0081	+0.0039	+0.0026
Timing jitter, ± 0.9 s	-0.0003	+0.0003	+0.0006	+0.0003
Burst dropout, 1 h	+0.0050	+0.0209	+0.0239	-0.0093
Voltage spikes, ± 0.20 V	+0.0000	-0.0003	+0.0002	-0.0001

Table B.7: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

Estimator	Family balanced	Equal scenario	Highest level plus offsets
DM	0.562	0.574	0.615
HDM	0.468	0.504	0.512
HECM	0.428	0.560	0.576
DD	0.842	0.781	0.665

Table B.8: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means on the common source-sample interval.

Metric	DM	HDM	HECM	DD
Initial recovery/censor time [h]	22.28	17.21	1.12	1.29
Recovery excess-error AUC [SOC h]	1.735	1.738	0.078	0.116
Right-censored fraction	0.89	0.67	0.00	0.00
Relapse fraction after first hold	0.00	0.22	0.17	0.78
Median SOC-core latency [μ s]	0.171	0.165	23.9	424.9 ^a
Flash [KiB]	27.0	27.1	470.1	115.9 ^a
Peak runtime RAM [KiB]	2.4	2.4	2.5	4.1 ^a

^aContinuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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753 The author contribution statement is provided in the non-blinded submission
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755 **Declaration of competing interest**

756 The authors declare that they have no known competing financial interests or
757 personal relationships that could have appeared to influence the work reported
758 in this paper.

759 **Declaration of generative AI and AI-assisted technologies**
760 **in the manuscript preparation process**

761 During the preparation of this work the author(s) used ChatGPT, DeepL,
762 and Grammarly in order to improve the linguistic quality and style of the
763 manuscript. After using these tools, the author(s) reviewed and edited the con-
764 tent as needed and take(s) full responsibility for the content of the published
765 article.

766 **Data availability**

767 The dataset is publicly available through its cited TU Berlin repository
768 record [28]. Code and results are maintained in a public GitHub repository.
769 A versioned release will archive the fixed experimental definitions and random
770 seeds, trained model parameters, input scalers, ECM parameterization, run

771 summaries, analysis and plotting procedures, and software environment needed
772 to reproduce all reported results.

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